

Nocebo effects are stronger and more persistent than placebo effects in healthy individuals

Reviewed Preprint

v2 • October 9, 2025

Revised by authors

Reviewed Preprint

v1 • April 22, 2025

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eLife Assessment

In this preregistered study, Kunkel and colleagues set out to compare the magnitude and duration of placebo versus nocebo effects in healthy volunteers, and also to examine the different factors contributing to these effects. The authors follow a rigorous methodology in a within-subjects design, taking into consideration standard conventions for manipulation of expectations, and using an appropriate sham condition. They present **compelling** evidence of long-lasting placebo and nocebo effects, with nocebo responses demonstrating consistently greater strength. These **valuable** results have the potential for a great impact in the field of experimental and clinical pain.

<https://doi.org/10.7554/eLife.105753.2.sa3>

Abstract

Placebo and nocebo effects illustrate the profound influence of cognitive-affective processes on symptom perception and treatment outcomes, with the potential to significantly alter responses to medical interventions. Despite their clinical relevance, the question of how placebo and nocebo effects differ in strength and duration remains largely unexplored. Using a within-subject design in 104 healthy individuals, we investigated and directly compared the magnitude and persistence of placebo and nocebo effects on experimental pain. Effects were assessed immediately after their induction through verbal instructions and conditioning and at a one-week follow-up. The study was preregistered in the German Clinical Trials Register (registration number: DRKS00029228). Significant placebo and nocebo effects were detected on day 1 and day 8, but nocebo effects were stronger on both test days. Sustained effects after one week were primarily predicted by individuals' experienced effects on day 1. Our findings underscore the enduring nature of placebo and nocebo effects in pain, with nocebo responses demonstrating consistently greater strength, which is consistent with an evolutionarily advantageous 'better-safe-than-sorry' strategy. These insights emphasise the significant

impact of nocebo effects and stress the need to prioritise efforts to mitigate them in clinical practice.

Introduction

Placebo and nocebo effects are intriguing phenomena that have generated considerable research interest in medicine, psychology, and neuroscience^{1–5}. Belief in the effectiveness or ineffectiveness of a treatment can reduce or increase symptoms, highlighting the powerful interaction between perception, physiology and cognitive-affective processes. Harnessing the power of positive expectations could complement standard medical treatments, and thereby enhance overall treatment outcome^{6,7}. Conversely, awareness of nocebo effects is important to minimise negative expectations and side effects in clinical practice^{3,8}. Moreover, it is relevant in placebo-controlled clinical trials where nocebo effects, manifesting as adverse events in the placebo group, can decrease treatment adherence and even lead to treatment discontinuation⁹. Recent insights into both phenomena have therefore led to a growing call to systematically utilize placebo effects and to learn to avoid nocebo effects in clinical care.

While extensive investigations have focused on the psychological and neurobiological mechanisms underlying positive expectations and their effect on symptom perception^{2,7}, our understanding of negative expectations and nocebo effects is comparably sparse despite evidence that nocebo effects can be moderate to large in size¹⁰. Even less is known about the longevity of the effect, a crucial factor for assessing its impact on treatment outcome in real life scenarios.

Importantly, there is evidence suggesting that an individual's susceptibility to nocebo information may not simply mirror their capacity for placebo analgesia. Early research by Colloca et al.¹¹ demonstrated that a single session using non-painful stimuli induced a nocebo effect but failed to elicit a placebo effect, indicating that negative expectations may be more readily triggered than positive ones. Moreover, nocebo effects seem to generalise more easily to other symptoms or treatments^{12,13}. Given the evolutionary relevance of anticipating negative, threatening and potentially harmful events it seems reasonable to assume that negative expectation and its effect on health outcome is an integral aspect of promoting safety behaviours and are thus more persistent than positive expectation. To accurately gauge an individual's capacity to produce placebo and nocebo effects and compare their magnitude and duration, it is essential to investigate both effects within the same individual.

Here we investigated immediate and sustained effects of positive and negative treatment expectations on experimentally induced heat pain in N= 104 healthy volunteers. Our experimental approach allowed for the trial-by-trial modulation of expectations for pain relief and pain aggravation in a within-subject design. Verbal instructions were combined with conditioning along with a sham electrical stimulation, which was introduced to participants as a method to 'induce frequency-dependent changes in pain sensitivity'. Treatment expectations and pain perception of physically identical medium-level heat stimuli were assessed immediately after expectancy induction (day 1), but also one week later (day 8) to investigate the longevity of both placebo analgesia and nocebo hyperalgesia. We also assessed psychological variables to explore whether they modulate or predict an individual's susceptibility, effects, and persistence of expectancy effects on pain. We hypothesized that negative expectations and nocebo effects would be stronger than placebo effects induced on day 1, and that negative expectations and their effects are more resistant to extinction and would therefore still be stronger on day 8.

Our data confirm that, although significant placebo and nocebo effects were found on days 1 and 8, the nocebo effect was consistently stronger. Both effects were primarily influenced by the most recent experience of pain reduction and pain increase but were also susceptible to psychological factors.

Results

The calibration procedure determined one temperature level for the placebo condition and one for the nocebo condition that were equidistant from the control condition. These temperatures were used in the conditioning procedure to induce the perception of pain reduction (placebo hyperalgesia) and pain aggravation (nocebo hyperalgesia), respectively (for details see [Fig. 1](#) and Supplementary Results). In the test sessions on days 1 and 8, however, the same medium-level temperature of the control condition was applied in all three conditions. The analyses include comparisons between all three conditions (i.e., placebo, nocebo and control) and comparisons between placebo effects (i.e., control vs. placebo) and nocebo effects (i.e., nocebo vs. control). To identify variables associated with placebo or nocebo effects on day 1 or day 8, we conducted multiple regression analyses including expected and experienced effects as well as psychological variables as potential predictors.

Placebo and nocebo effects on day 1

The comparison of pain intensity ratings acquired after the conditioned expectancy manipulation in the first test session on day 1 confirmed differences between three conditions ($F(1.28, 131.96) = 96.32$, $p < .001$) with both a significant placebo effect (control vs. placebo condition: $t(103) = 3.92$; $p < .001$; 95% CI, 2.07 to 6.32; $d = 0.38$) and a significant nocebo effect (nocebo vs. control condition: $t(103) = 14.88$; $p < .001$; 95% CI, 9.78 to 12.79; $d = 1.46$; [Fig. 2A](#)). A direct comparison of both effects revealed a stronger nocebo effect than placebo effect (nocebo effect: $M = 11.29$, $SD = 7.73$; placebo effect: $M = 4.19$, $SD = 10.92$; $t(103) = 6.56$; $p < .001$; 95% CI, 4.95 to 9.24; $d = .64$; [Fig. 2B](#)).

Placebo and nocebo effects on day 8

In the second test session, seven days after the expectancy manipulation, pain intensity ratings remained to be different between conditions ($F(1.58, 153.34) = 111.93$, $p < .001$), despite the same stimulation intensity. Participants still showed a significant placebo effect (control vs. placebo condition: $t(97) = 6.06$; $p < .001$; 95% CI, 3.08 to 6.09; $d = .61$) as well as a nocebo effect ($t(97) = 10.79$, $p < .001$; 95% CI, 7.29 to 10.58; $d = 1.09$). As on day 1, a direct comparison between both effects using difference scores showed a stronger nocebo than placebo effect on day 8 (nocebo effect: $M = 8.93$, $SD = 8.20$; placebo effect: $M = 4.58$, $SD = 7.50$; $t(97) = 3.90$, $p < .001$, 95% CI, 2.14 to 6.56; $d = .39$) ([Fig. 2B](#)).

Comparison of day 1 and day 8

A direct comparison of placebo and nocebo effects on day 1 and day 8 pain intensity ratings showed a main effect of *Effect* with a stronger nocebo effect ($F(1,97) = 53.93$, $p < .001$, $\eta^2 = .36$) but no main effect of *Session* ($F(1,97) = 2.94$, $p = .089$, $\eta^2 = .029$). The significant *Effect* $\times$ *Session* interaction indicated that the placebo effect and the nocebo effect developed differently over time ($F(1,97) = 3.98$, $p = .049$, $\eta^2 = .039$). While the nocebo effect decreased significantly from day 1 to day 8 ($t(97) = 2.68$, $p = .018$, 95% CI, 0.66 to 4.44; $d = 0.27$), the placebo effect did not change ($t(97) = -0.517$; $p = .606$; 95% CI, -2.47 to 1.45, $d = -0.05$), possibly due to the lower starting point on day 1. Of note, placebo and nocebo effects were significantly positively correlated at day 1 ($r = 0.34$; $p < .001$) but showed no significant relationship on day 8 ($r = 0.01$; $p = .903$).

Evolution of differences between placebo and nocebo effects

To test whether the difference between the placebo and the nocebo condition already evolved during conditioning, we first compared pain intensity ratings provided during conditioning where stimulus intensities were manipulated unbeknownst to the participant. As intended, heat stimuli

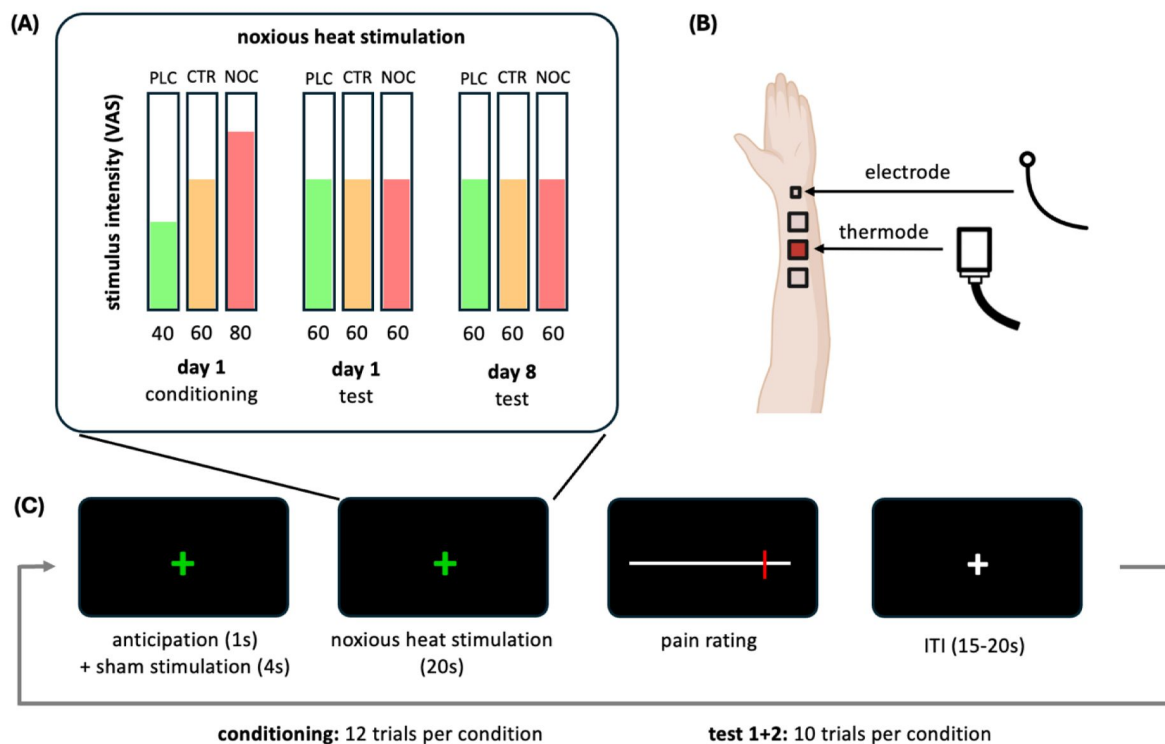


Figure 1

Study and trial design.

(A) Study design: On day 1, participants underwent a conditioning procedure in which a noxious heat was applied directly after a (sham) TENS stimulation in three conditions. In the placebo condition (PLC), the thermal stimulation was lowered to VAS 40, in the nocebo condition (NOC), it was increased to VAS 80 and in the control condition (CTR) it remained unchanged (VAS 60). During the two tests on day 1 and day 8, the same moderate stimulation intensity of VAS 60 was applied in all three conditions. (B) Position of the electrode on the inner lower left arm for (sham) TENS stimulation (approximately 2.5 cm above the wrist) and the thermode at three possible locations (approximately, 3.5 cm above the electrode with a distance of 0.5 cm between each of the three locations). (C) Trial design: Following the presentation of a visual cue to indicate the condition (e.g., green cross for the placebo condition), first the sham TENS stimulation and then the heat stimulus were applied before participants rated the pain intensity on a visual analogue scale.

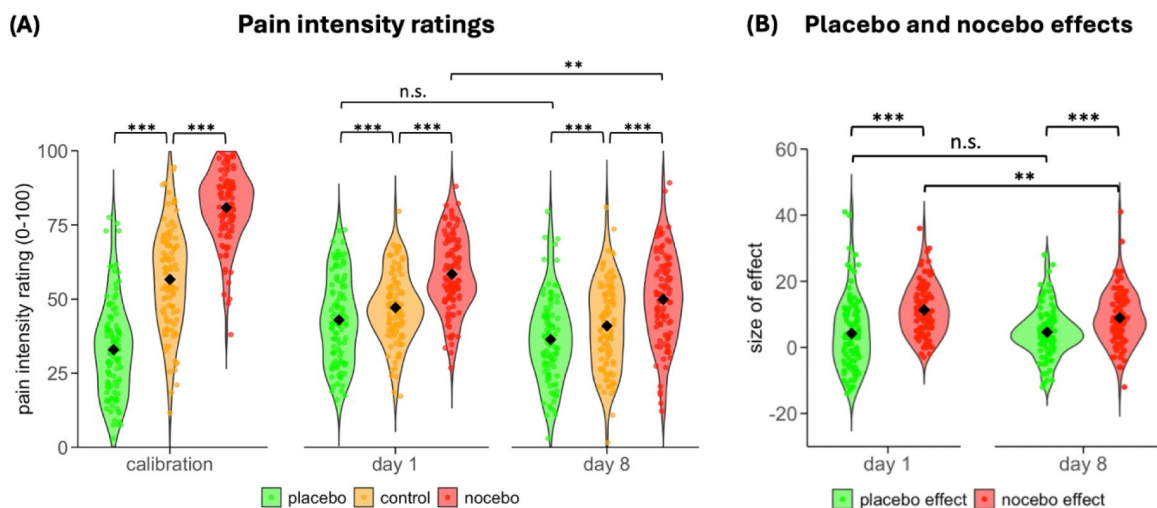


Figure 2

Pain intensity ratings and placebo and nocebo effects during calibration and test sessions.

(A) Mean pain intensity ratings in the placebo, nocebo and control condition during calibration, and during the test sessions at day 1 and day 8. (B) Placebo effect (control condition - placebo condition, i.e., positive value of difference) and nocebo effect (nocebo condition - control condition, i.e., positive value of difference) on day 1 and day 8. Error bars indicate the standard error of the mean, circles indicate mean ratings of individual participants. ***: $p < .001$, **: $p < .01$, n.s.: non-significant.

applied during placebo conditioning were rated as less painful than stimuli applied in the control condition (control vs. placebo condition: $t(103) = 20.56$; $p < .001$; 95% CI, 20.98 to 25.45; $d = 2.02$). Similarly, stimuli applied during nocebo conditioning were rated as more intense than stimuli in the control condition: $t(103) = 33.42$; $p < .001$; 95% CI, 31.16 to 35.09; $d = 3.28$) (**Fig. 3A**). However, the pain ratings revealed a stronger conditioning effect for the nocebo condition than the placebo condition (pain worsening effect: $M = 33.12$, $SD = 10.11$, pain relief effect: 23.21 , $SD = 11.51$; $t(103) = 5.96$, $p < .001$, 95% CI, 6.61 to 13.20; $d = .59$, **Fig. 3B**).

To explore the formation of the pain relief and pain worsening during conditioning in more detail, we compared changes in trial-by-trial pain intensity ratings over the conditioning phase between the three conditions (**Fig. 3C**). This analysis showed no significant main effect of *Trial* ($F(4.37, 341.01) = 1.25$, $p = .289$, $\eta^2 = 0.016$), indicating that there was no general change in ratings over time. However, as shown by a significant main effect of *Condition* ($F(1.84, 143.76) = 950.85$, $p < .001$, $\eta^2 = 0.924$) and more importantly a significant interaction between *Trial* and *Condition* ($F(13.93, 1086.45) = 4.93$, $p < .001$, $\eta^2 = 0.059$), changes in ratings over time differed between the three conditions. Separate regression analyses for each condition showed that although ratings decreased in the placebo condition ($\beta = -0.22$), the decrease was not significant ($p = .242$). Conversely, both the nocebo and the control condition showed an increase in ratings over time, but the increase only reached significance in the nocebo condition ($\beta = 0.39$, $p = .048$; control condition: $\beta = 0.09$, $p = .512$) indicating a stronger formation of nocebo hyperalgesia already during conditioning, despite rigorous calibration to intensities equidistant from the control condition.

To test whether the differences between placebo effects and nocebo effects on day 1 and day 8 could be explained by stronger nocebo conditioning, we repeated the previous comparisons between both effects, but this time included the difference in conditioning (nocebo condition minus placebo condition) as a covariate. While the difference in conditioning could indeed explain a significant part of the variance ($F(1, 102) = 5.85$, $p = .017$, $\eta^2 = 0.054$), the nocebo effect was still significantly stronger on day 1 (main effect *Effect*: $F(1, 102) = 20.79$, $p < .001$, $\eta^2 = 0.169$), indicating genuine differences in the underlying mechanisms and temporal dynamics. A similar (albeit weaker) result was found for day 8 with a significant difference between the placebo and the nocebo effect (main effect *Effect*: $F(1, 96) = 4.81$, $p = .031$, $\eta^2 = 0.048$) in addition to a significant effect of the difference in conditioning ($F(1, 96) = 4.38$, $p = .039$, $\eta^2 = 0.044$).

Expectancy ratings

Given the proposed key role of expectations in placebo and nocebo effects, we also obtained expectancy ratings prior to each testing session. Because expectancy ratings were not normally distributed, we used a non-parametric analysis approach. Expectations that the pain would improve in the placebo condition and worsen in the nocebo condition did not differ significantly before conditioning, confirming that our verbal instruction had induced equally strong expectations ($Z(104) = -0.34$, $p = .737$; **Fig. 4**). The conditioning procedure on day 1 significantly increased the expected pain relief (placebo) ($Z(104) = -3.76$, $p < .001$) but not the expected pain worsening (nocebo) ($Z(104) = -1.09$, $p = .556$) and a direct comparison showed significantly stronger placebo than nocebo expectations ($Z(104) = -2.71$, $p = .007$). Between day 1 and day 8, placebo expectations decreased significantly ($Z(98) = -3.09$, $p = .004$) and were no longer different from ratings before conditioning ($Z(104) = -0.96$, $p = 0.338$). Nocebo expectations also decreased ($Z(98) = -3.90$, $p < .001$) and were even significantly lower than before conditioning ($Z(98) = -2.30$, $p = .021$). As on day 1, the expected pain relief was significantly stronger than the expected worsening of pain ($Z(98) = -3.39$, $p = .001$).

Neither placebo nor nocebo expectations were significantly linked to the experienced effect on day 1 (placebo: Spearman's $\rho(104) = 0.10$, $p = .335$; nocebo: Spearman's $\rho(104) = 0.17$, $p = .093$) or day 8 (placebo: Spearman's $\rho(98) = 0.13$, $p = .187$; nocebo: Spearman's $\rho(98) = 0.88$, $p = .396$).

Figure 3

Mean and trial-by-trial pain intensity ratings, placebo and nocebo effects during conditioning.

(A) Mean pain intensity ratings of the placebo, nocebo and control condition during conditioning. (B) Placebo effect (control condition - placebo condition, i.e., positive value of difference) and nocebo effect (nocebo condition - control condition, i.e., positive value of difference) during conditioning. (C) Trial-by-trial pain intensity ratings (with confidence intervals) during conditioning. Error bars indicate the standard error of the mean, circles indicate mean ratings of individual participants. ***: $p < .001$.

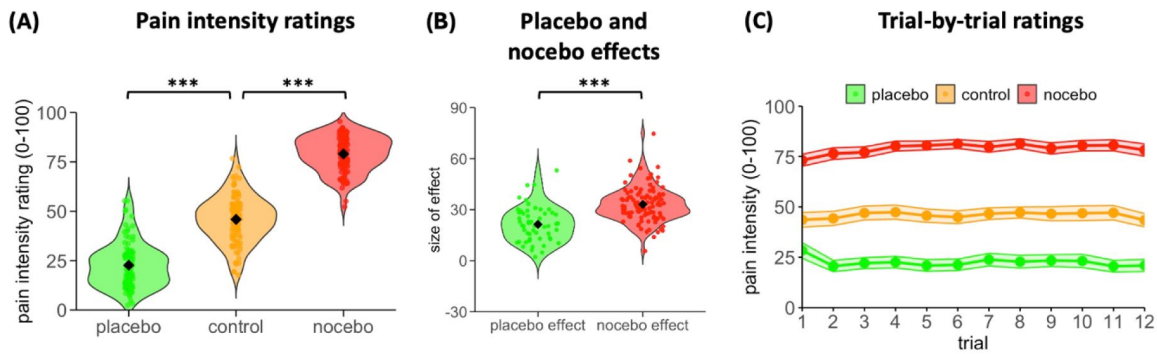
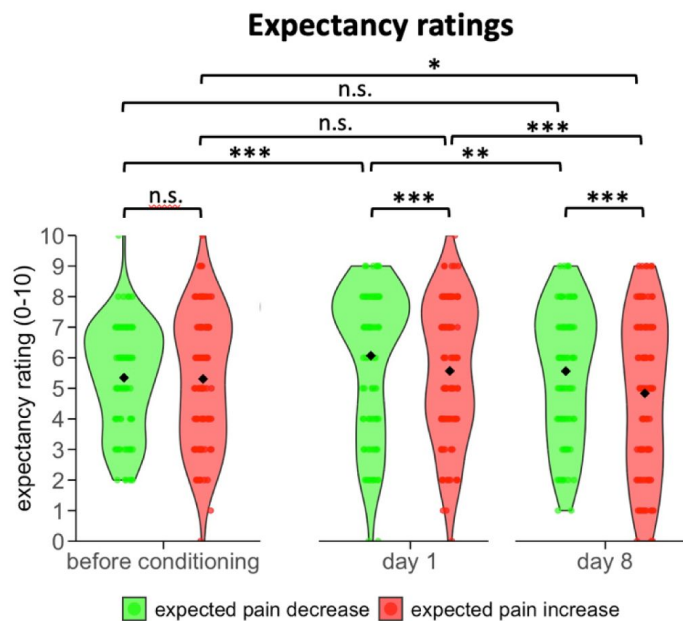


Figure 4

Expectancy ratings obtained before conditioning and before the test sessions on day 1 and day 8.

Expectations were assessed using the *Generic Rating Scale for Previous Treatment Experiences, Treatment Expectations, and Treatment Effects* (GEEE, Rief et al., 2021 [\[1\]](#)). The expected pain relief was derived from the item asking how much improvement the participant expected from the treatment on a 10-point Likert-scale from 0 (= no improvement) to 10 (= greatest improvement imaginable). Analogously, the expected pain increase (nocebo effect) was taken from the item asking how much worsening of pain they expected from the treatment from 0 (= no worsening) to 10 (= greatest worsening imaginable). Black diamond shapes indicate the mean and circles the individual scores. ***: $p < .001$, **: $p < .01$, * $p < .05$, n.s.: non-significant



Multiple linear regression analyses (expected and experienced effects)

Next, we employed multiple linear regression analyses to investigate the significance of expected (GEE ratings) and experienced placebo and nocebo effects (VAS ratings) for subsequent effects on both test days. Overall, the regression model for the placebo effect on day 1 explained 9.7% of the variance (Supplementary Table S1). The only predictive variable for the placebo response on day 1 was the pain relief during conditioning. In the equivalent model for the nocebo effect, none of the variables could significantly predict the nocebo response on day 1.

The regression model for the placebo effect on day 8 explained a total of 25.1% of the variance with two significant predictors: the placebo effect on day 1 and the placebo expectation on day 8 (Supplementary Table S1). For the nocebo response on day 8, the tested model explained 7.1% of the variance with the nocebo effect at day 1 as the only significant predictor (Supplementary Table S1). Together, these differences in the contribution of expectations and experienced effects between the placebo and the nocebo condition further substantiate that both effects are driven by different mechanisms.

Multiple regression analyses (expected and experienced effects plus psychological variables)

In the final analysis step, we tested whether psychological variables that have been linked to placebo and nocebo effects in the past, such as trait anxiety¹⁵, practitioner characteristics¹⁶, or somatosensory amplification¹⁷ could increase the predictive power of the previously tested models. On day 1, in addition to the significant prediction from the experienced conditioning effect that had already been significant in the previous model, somatosensory amplification emerged as a negative predictor of the placebo effect, indicating that individuals with a higher tendency for somatosensory amplification were less likely to experience placebo analgesia. The total variance explained in this model was 14.5% (Supplementary Table S2). This influence of somatosensory amplification was no longer detectable on day 8 where only the experienced placebo effect on day 1 and placebo expectations on day 8 were significant predictors but none of the psychological variables (total amount of variance explained: 26.4%).

The equivalent analyses for the nocebo effect revealed that higher nocebo effects were found when participants had rated the experimenter competence as high (Supplementary Table S2), pointing towards a potential iatrogenic effect of experimenter when they implied that pain could become worse with the treatment. The total amount of variance explained by this model was 10.6%. As for the placebo effect, none of the psychological variables predicted the nocebo effect on day 8. The total variance explained by this model with only the perceived nocebo effect on day 1 as a significant predictor was 1.6%.

Discussion

In this pre-registered, experimental study in healthy individuals, we investigated placebo analgesic and nocebo hyperalgesic effects immediately after a conditioned expectancy manipulation and seven days later. Three key findings emerged from our investigation. First, medium-to-large scale placebo and nocebo effects were found not only on day 1 but also one week later. Second, nocebo effects were consistently stronger than placebo effects, including during the conditioning phase, despite analogous conditioning protocols in both conditions. Third, placebo and nocebo effects are primarily driven by the most recent experience of these effects but were also susceptible to some psychological factors.

Sustained placebo and nocebo effects

While placebo effects have been shown to persist for an extended period of time after they have been induced, there are only a few studies that have investigated the longevity of nocebo effects so far and these studies focused on sustained effects within the same test session^{11,18}. In our study, nocebo effects were not only sustained over the period of a week, but they were also significantly stronger than the placebo effect on both test days (**Fig. 2B**). This finding aligns with broader evidence from learning studies, which demonstrate a greater influence of negative information on sensory perception^{19–21}, as well as similar effects observed in placebo and nocebo trials. For example, nocebo hyperalgesia was more easily induced via instructions than placebo analgesia²² and tended to extinguish more slowly^{18,22}. Additionally, in an experimental study involving healthy individuals, Colloca and colleagues¹¹ found that one session of conditioning was sufficient to induce a nocebo effect but not a placebo effect.

Stronger and more sustained nocebo effects are likely to be the result of a combination of different factors. Evolutionary psychology suggests that humans may have evolved to be more attuned to potential threats for survival. Negative information or expectations about harm may have carried more evolutionary significance, making individuals more sensitive to nocebo suggestions, a tendency often referred to as ‘better safe than sorry’. Confirmation for this assumption comes from brain imaging studies demonstrating a cognitive bias in which the brain tends to process negative information more readily than positive information. Moreover, negative expectations and fear tend to amplify sensory perception^{19–21}. When individuals anticipate negative outcome, their attention is often heightened which makes them susceptible to perceiving symptoms, even in the absence of an actual stimulus. In line with this assumption, nocebo effects have been shown to lead to anticipatory anxiety and autonomic arousal which mediated the effect on extinction in an experimental learning model¹⁸. It may be argued that the dominant nocebo effect observed in our study is the result of the stronger conditioning in the nocebo condition (**Fig. 3**). This asymmetry is noteworthy in and of itself because it occurred despite the equidistant stimulus calibration relative to the control condition prior to conditioning. It may be the result of different physiological effects of the stimuli over time or amplified learning in the nocebo condition, consistent with its heightened biological relevance, but it could also be a stronger effect of the verbal instructions in this condition. Importantly, the stronger nocebo effect observed on both test days remained significant even after accounting for the asymmetric conditioning effect, ruling out that conditioning differences alone explain the stronger nocebo effects. Instead, it suggests that the two effects may be induced and maintained by at least in part distinct mechanisms and temporal dynamics. Recent work using a predictive coding framework further suggests that nocebo effects may be less susceptible to prediction error than placebo effects²³, which could contribute to their greater persistence and strength in our study. This is supported by the observation that, similarly to a previous study¹¹, a significant correlation between placebo and nocebo effects was found on day 1 but was no longer detectable at the follow-up one week later. Interestingly, our expectancy manipulation increased placebo expectations, but had no significant effect on nocebo expectations (**Fig. 4**). Furthermore, expectations were not correlated with actual placebo or nocebo effects on either test day. While this may seem surprising, it has recently been suggested that these correlations depend on whether expectations are measured in the same format as the pain experience or as a difference measure, as in our study²⁴. Further research is therefore needed to investigate the effects of assessment methods on such associations.

It is important to note that our study was designed in alignment with previous studies addressing similar questions (e.g.,²⁵). Our primary aim was to directly compare placebo and nocebo effects in a within-subject design and assess their persistence of these effects one week following the first test session. One limitation of our approach is the relatively short duration of each session, which

may have limited our ability to examine the trajectory of responses within a single session. Future studies could address this limitation by increasing the number of trials for a more comprehensive analysis.

Past effects predict future effects

To explore the relative influence of expectations and prior experience in more detail, we conducted separate regression analyses for placebo and nocebo effects on both test days, using expectations and perceived effects as predictors. The analyses revealed that experienced pain reduction and increase were significant predictors of subsequent effects, especially for the placebo effect on day 1 and day 8, and for the nocebo effect on day 8 (Supplementary Table S1). This highlights the strong impact of sensory experience on subsequent effects, in line with studies on learning²⁶, meta-analyses of behavioural placebo analgesia²⁷, and previous studies on carry-over effects between analgesic treatments¹². Notably, the most recent experience was the most predictive in all three analyses; for instance, the placebo effect on day 8 was predicted by the placebo effect on day 1, not by the initial conditioning. This finding supports the Bayesian inference framework, where recent experiences are weighted more heavily in the process of model updating because they are more likely to reflect the current state of the environment, providing the most relevant and immediate information needed to guide future actions and predictions²⁸. Interestingly, while a change in pain predicted subsequent nocebo effects, it seemed less influential than for placebo effects. This aligns with findings that longer conditioning enhanced placebo effects, while it did not affect nocebo responses¹¹ and the conclusion that nocebo instruction may be sufficient to trigger nocebo responses. Using Bayesian modeling, future studies could identify individual differences in the development of placebo and nocebo effects by integrating prior experiences and sensory inputs, providing a probabilistic framework for understanding the underlying mechanisms.

The role of psychological variables in immediate and sustained placebo and nocebo effects

Our extended regression models, incorporating psychological variables, highlight two interesting predictors: somatosensory amplification and perceived practitioner competence (Supplementary Table S2). Somatosensory amplification, described as a tendency to experience bodily symptoms as intense, noxious and disturbing¹⁷, was associated with a weaker placebo effect on day 1. This may be due to higher-level evaluative processes²⁹, leading individuals to perceive symptoms as more threatening, which in turn diminishes the influence of cognitive processes that typically drive placebo effects. Additionally, our study suggests that nocebo effects can be linked to the perceived competence of the experimenter. While practitioner competence - alongside perceived warmth - usually enhances positive treatment expectations³⁰ and treatment outcome^{16,31}, it might also make negative suggestions more convincing and thereby amplify nocebo responses through increased anxiety or hypervigilance. This finding underscores the dual-edged nature of competence in patient-practitioner interactions, where heightened credibility could inadvertently strengthen nocebo effects.

Our findings have important implications for clinical research and practice. First, they underscore the necessity of prolonged observation periods in clinical trials to accurately capture the durability of these effects. Second, they emphasise the importance of not dismissing early signs of nocebo effects as they may persist and undermine otherwise treatments if left unaddressed. Third, our findings advocate for a stronger focus on the prevention of nocebo effects. While considerable effort has been made to leverage placebo effects, it is equally - if not more - crucial to focus on minimising nocebo effects, which seem to be triggered more easily. Fortunately, nocebo effects can often be avoided by adopting simple, effective strategies to improve patient-practitioner communication. For example, positive framing, avoiding unnecessary focus on potential side effects, or building a trusting relationship can reduce the likelihood of triggering nocebo effects. In

a time where cost-effectiveness is paramount, and healthcare resources must be carefully allocated, prioritizing the prevention of nocebo effects should be a key strategy to enhance treatment outcome and reduce overall healthcare costs.

In summary, our findings indicate that nocebo effects are indeed more than the flipside of a placebo effect and that the two phenomena may be sustained by distinct mechanisms. These insights shed light on the factors that exacerbate nocebo effects and underscore the importance of carefully managing communication in clinical and experimental settings.

Materials and methods

Participants

As there are no studies investigating within-subject effects that could provide indications for the relevant effect sizes, the study was powered to detect small effect sizes for the hypothesized differences between the placebo and nocebo condition ($d = 0.2 - 0.25$, $\alpha = 0.05$, power = 0.95, $N = 84120$). A total of $N = 112$ healthy volunteers were recruited through public adverts and received structured telephone interviews for screening purposes. Exclusion criteria comprised red-green colour blindness, drug use in the last four weeks, alcohol consumption in the last 24 hours, caffeine consumption on the test day, acute or chronic pain, a history of or acute psychiatric disorders (including major depression, schizophrenia and suicidality), hypersensitivity or other neurological diseases, acute infections, skin diseases, surgical procedure under anaesthesia in the last six months, use of analgesic or anticoagulant medications within the last 24 hours, use of any other medication in the last 7 days (except thyroid medication, hormonal contraceptives or allergy medication), pregnancy or breastfeeding. People were also ineligible if they had taken part in another study using electrical stimulation or experimental heat pain in the last six months before the study. Eight participants were excluded on the first testing day, two because of technical problems, two because they did not meet the inclusion criteria (due to caffeine consumption and yellow fever vaccination), and four showed a low or inconsistent pain sensitivity rendering the experimental manipulation ineffective (e.g., 80% of the pain stimuli were rated with a VAS score of 0). The final sample for the analyses of day 1 consisted of 104 participants (63 female and 41 male, mean $\pm$ SD age: 24.92 ± 3.47 , range = 18 to 36 years). Six participants were unable to take part in the follow-up examination on day 8 for the following reasons: one due to personal illness, two because of the experimenter's illness, one failed to attend, another participated in a similar experiment between sessions, and one took pain medication on day 8. As a result, the final sample for day 8 consisted of 98 participants (59 female, 39 male, mean age $\pm$ SD: 24.86 ± 3.29 years, range: 18 to 36 years). The study was preregistered with the German Clinical Trials Register (<https://drks.de/search/de/trial/DRKS00029228>; registration number: DRKS00029228). Ethics approval was granted by the University Hospital Essen (22-10597-BO). The experiment adhered to the principles outlined in the 2013 Declaration of Helsinki. Informed written consent was obtained from all participants, who received 120 Euros for their participation.

Study design and procedure

This study used a within-subjects design (Fig. 1) to investigate the immediate and sustained effects of three types of experimentally induced treatment expectations on heat pain perception: expectations of reduced pain (placebo condition), expectations of increased pain (nocebo condition) and expectations of no change in pain (control condition). The experiment was carried out on two days with an approximate duration of three hours on day 1 and one hour on day 8. On the first day (day 1), treatment expectations were induced using verbal instructions in combination with a conditioning procedure. During conditioning, participants learned to associate the presentation of one of three visual, differently coloured cues with a reduction of heat-induced pain through a (sham) 'transcutaneous electrical nerve stimulation (TENS) device' that was introduced as an analgesic treatment in the placebo condition. A second cue signalled an increase

in pain the placebo condition and the third cue signalled no change in pain in the control condition. As in previous studies using conditioning to induce placebo and placebo effects^{11,32,34}, unbeknownst to the participant the heat stimulation was reduced from VAS 60 to VAS 40 in the placebo condition, increased to VAS 80 in the placebo condition and left unchanged at VAS 60 in the control condition. In the subsequent first test session, the same moderate stimulation intensity of VAS 60 was used in all three conditions. To explore the longevity of the induced conditioned effects, participants underwent the same testing procedure but no conditioning a week later (day 8) with all three visual stimuli again followed by the same moderate temperature stimulation (VAS 60). Participants' condition-specific treatment expectations and trial-by-trial pain intensity ratings were recorded as outcome measures. The study also comprised structural and functional magnetic resonance imaging (MRI) that took place on a separate day before day 1 (methods and data on this part will be reported elsewhere).

During the experiment, the participants were seated in a chair in front of a computer in a behavioural laboratory setting with a keyboard as response device. The left arm was positioned on a long cushion resting on the table while the right hand operated the keyboard. The experimenter faced the participant from the opposite side of the table with the computer screen between them.

Presentation of visual stimuli, delivery of thermal and electrical stimuli, and outcome recording were implemented using Presentation (Version 22.0, Neurobehavioral Systems, Inc., Berkeley, CA).

Calibration of the noxious thermal stimulation

Heat stimuli were calibrated to each participant's level of sensitivity. First, we used the Method of Limits³⁵ to determine the individual heat pain threshold (HPT) in three consecutive trials. In the subsequent calibration procedure, participants rated 21 noxious heat stimuli with varying temperature levels around the HPT (-1°C - $+3.5^{\circ}\text{C}$) on a visual analogue scale (VAS) with endpoints 0 (= "not painful at all") and 100 (= "unbearably painful"). These ratings were entered into a linear regression ($\ln(\text{VAS rating} \sim \text{temperature})$) in RStudio (except for the first rating due to familiarisation effects) to determine the temperature levels rated as VAS 40, 60 and 80. These temperatures were applied twice in a short subsequent test to ensure that the calculated heat levels induced the intended pain intensity. The 20-second contact heat pain stimuli were delivered using a Pathway advanced thermal stimulator with a 30×30 mm activation area (Pathway System, Medoc, Israel). The thermode was attached to one of three possible locations on the medial inner aspect of the left forearm using a tourniquet, maintaining a standardised distance of 3.5 cm from the electrode maintained via a template. To prevent sensitization or habituation, the three different stimulation sites were used. The thermode was moved to another of the three locations after calibration and conditioning, following a fixed, pseudorandomised order.

(Sham) transcutaneous electrical nerve stimulation

Participants were instructed that the applied non-painful electrical stimulation with different frequencies would either increase, decrease, or not influence pain perception, respectively. The electrical stimuli were applied to the left volar forearm approximately 2.5 cm proximal of the wrist using a Digitimer stimulator (Welwyn Garden City, England, model DS7A) that was connected to a surface electrode (Specialty Developments, Bexley, UK) with a diameter of approximately 5mm attached to the skin using medical tape. During calibration, the initial stimulation intensity for the four-second train of stimuli started at 0.9 mA and increased in increments of 0.1 mA until participants noticed a clear but non-painful sensation. This intensity was then tested by applying four four-second stimuli. If participants rated at least 2/4 of the stimuli between 25 and 35 on a VAS from 0-100, this final stimulation intensity was carried forward to be used throughout the test sessions. If the electrostimulation was not perceivable on day 8, the calibration was repeated once more before the start of the other experiments.

Conditioning procedure

During the conditioning session, participants' expectations of pain relief and pain increase were modulated using verbal instructions and electrical stimulation coupled with coloured visual cues. Specifically, participants were told that the electrical stimulation would either increase (nocebo instruction), decrease pain (placebo instruction), or have no influence on their pain perception (control instruction) depending on the frequency of the stimulation. The direction of change would be indicated by the colour of a cross that was shown in the centre of the computer screen. A green cross indicated a decrease in pain (placebo condition), a red cross indicated an increase of pain (nocebo condition) and a yellow indicated no change (control condition). In fact, unbeknownst to the participants, in placebo trials the green cross was followed by low-intensity heat stimulation calibrated at VAS 40 to induce a sense of pain reduction through the electrical stimulation, whereas the red cross was followed by a high-intensity heat pain calibrated at VAS 80 for a sense of pain increase (**Fig. 1**). In control trials the yellow cross was followed by a VAS 60 heat pain stimulus. This temperature manipulation was applied to all trials of the conditioning procedure, respectively. The order of condition was pseudorandomised, and each trial type was repeated twelve times during the conditioning procedure. Due to a randomisation error, 25 participants received an unbalanced number of trials per condition (i.e., 10 × VAS 40, 14 × VAS 60, 12 × VAS 80). However, mean pain intensity ratings during the conditioning phase did not differ significantly between these participants and the remaining sample in any of the three conditions (2-sample t-test (2-sided); placebo condition: $t(102) = -0.806$, $p = .422$, nocebo condition: $t(102) = 0.849$, $p = .398$, control condition: $t(102) = 0.390$, $p = .697$).

Test sessions

Placebo and nocebo responses were assessed during both test sessions on day 1 and day 8 following the same procedure as the conditioning session, but without temperature manipulation. Instead, the same target temperature corresponding to VAS 60 was maintained across all conditions (see **Fig. 1** for details of the design). The order of condition was pseudorandomised, and each trial type was repeated ten times in each test session. On day 8, one stimulus per stimulation intensity (i.e., VAS 40, 60 and 80) was applied before the start of the test session to re-familiarise participants with the thermal stimulation. Note that participants were informed that these pre-test stimuli were part of the recalibration and refamiliarization procedure conducted prior to the second test session.

Pain intensity ratings

During the conditioning and the test sessions, participants provided pain intensity ratings on a VAS with endpoints points 0 (= “not painful at all”) and 100 (= “unbearably painful”). The cursor was positioned randomly on the scale at the beginning of the rating period. Participants could move the cursor by pressing the left or right arrow key and were asked to confirm their rating with the ‘enter’ key (no time limit).

Reaction time task

During the conditioning and test sessions, a reaction time task was included at the beginning of each trial in which participants had to respond as quickly as possible to a target stimulus (a blue cross) by pressing the left arrow key to ensure sustained attention. The blue cross appeared for 300 ms with a jittered onset at the beginning of each trial, i.e., 0 to 5 s after trial onset.

Psychological questionnaires

Before calibration on day 1, participants completed the German version of the following questionnaires using an online survey system (LimeSurvey, LimeSurvey GmbH, Hamburg, Germany): the Generic Rating for Treatment Pre-Experiences, Treatment Expectations, and Treatment Effects (GEEE¹⁴), the Somatosensory Amplification Scale (SSAS^{17,36}), the Perceived Stress Scale (PSS-10^{37,38}), anxiety and depression (STADI Trait³⁹) and the Pain Catastrophizing Scale (PCS^{40,41}). Warmth and competence of the experimenter were assessed as described in Seewald & Rief³⁰ at the end of day 1. In short, participants were asked the question how the experimenter seemed to them and provided ratings on a 5-point scale ranging from 1 (= not at all) to 5 (= extremely) for the following descriptors in German: “friendly”, “well-intentioned”, “trustworthy”, “warm”, “good-natured” and “sincere” to capture experimenter warmth and “competent”, “confident”, “capable”, “efficient”, “intelligent” and “skilful” for experimenter competence. The mean across items of each scale was used in further analyses.

Treatment expectation ratings using the GEEE and the emotional state using STADI State were also collected before conditioning, after conditioning, and before test session 2 on day 8. Treatment effects were rated after conditioning and after test sessions 1 and 2. Note that participants also completed the following questionnaires as part of a larger project: Fear of Pain Questionnaire (FPQ-III), Behavioral Inhibition and Behavioral Activation (BIS/BAS) Scales, 10-item-Big-Five-Inventory (BFI-10) and the Positive and Negative Affect Schedule (PANAS). Responses to these questionnaires will be analysed elsewhere.

Statistical analyses

Data were analysed using SPSS (version 29.0.2.0). For each of the three conditions, mean pain intensity ratings for the calibration phase, conditioning phase and tests on day 1 and day 8 were calculated across the trials of the respective phase. Nocebo effects were defined as the difference in pain intensity ratings between the nocebo and the control condition (nocebo - control), placebo effects as the difference between the control and the placebo condition (control - placebo). Comparisons of stimulation intensities and pain intensity ratings between conditions were carried out using repeated-measures ANOVAs with the within-subject factor *Condition* (placebo, nocebo, control) followed by post hoc Bonferroni-corrected pairwise comparisons. To compare the magnitude and persistence of placebo and nocebo effects over time, an ANOVA with the within-subject factors *Effect* (placebo effect, nocebo effect) and *Session* (day 1, day 8) was used. The analysis of trial-by-trial ratings used an ANOVA with the within-subject factors *Condition* (placebo, nocebo, control) and *Trial* (trial 1 to 12). To account for interindividual differences in conditioning, the difference between the pain worsening and pain relief during the conditioning phase was entered as a covariate in the comparison of pain intensity ratings at day 1 and day 8 (ANCOVA). Degrees of freedom were corrected using the Greenhouse-Geisser estimate of sphericity. To explore the relationship between placebo and nocebo effects on both test days, we calculated the Pearson correlation coefficient. Because expectancy ratings were not normally distributed, non-parametric Wilcoxon signed rank tests were used to compare these ratings between conditions and timepoints and Spearman's rho was calculated for correlations between pain intensity and expectancy ratings. All questionnaires were analysed according to their respective manuals.

Separate multiple linear regression analyses were performed to examine the influence of expectations (GEEE ratings) and experienced effects (VAS ratings) on subsequent placebo and nocebo effects. For day 1, the placebo effect was entered as the dependent variable and the following variables as potential predictors: (i) expected pain improvement with placebo before conditioning, (ii) pain relief during conditioning and (iii) the expected pain improvement with placebo before the test session at day 1. The equivalent analysis was conducted for the nocebo

effect but with (i) expected pain worsening with placebo before conditioning, (ii) pain worsening during conditioning and (iii) the expected pain worsening with placebo before the test session at day 1 as predictors.

To predict placebo responses a week later ($VAS_{\text{control}} - VAS_{\text{placebo}}$ at day 8), the same independent variables were entered as for day 1 but with the following additional variables (i) pain ratings during test trials at day 1 and (ii) the expected pain improvement with placebo before the test session at day 8. In the equivalent analysis for the placebo effect on day 8 as dependent variable, we added (i) the placebo effect at day 1 and (ii) the expected pain worsening with placebo before the test session at day 8.

To explore whether psychological variables could explain additional variance in the regression analyses, we repeated all four analyses described above but included scores from these questionnaires as additional independent variables: Somatosensory Amplification Scale (SSAS), Perceived Stress Scale (PSS-10), trait anxiety and depression (STADI trait), Pain Catastrophizing Scale (PCS) and experimenter warmth and competence scores.

In all analyses, a significance level of $p < .05$ was used, and pairwise comparisons were conducted using two-tailed p-values. For all multiple regression analyses, the regression coefficient is reported.

Acknowledgements

The work was funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) - Project-ID 422744262 - TRR 289 (gefördert durch die Deutsche Forschungsgemeinschaft (DFG) - Projektnummer 422744262 - TRR 289).

Additional information

Author contributions

AK and KS contributed equally to this work as shared first authors. AK: Programming, Data acquisition, data analysis, writing, review & editing. KS: Conception, data analysis, interpretation, writing, review & editing. JS: data acquisition, writing. TS: data acquisition. HH: data curation, data analysis, interpretation, review & editing. KW: data analysis, interpretation, writing, review & editing. UB: funding acquisition, project administration, conceptualization, methodology, supervision, writing, review & editing

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Reviewer #1 (Public review):

Summary:

The study aimed to: (1) assess the magnitude of placebo and nocebo effects immediately after
induction through verbal instructions and conditioning, (2) examine the persistence of these

effects one week later, and (3) identify predictors of sustained placebo and nocebo responses over time.

Strengths:

An innovation was to use sham TENS stimulation as the expectation manipulation. This expectation manipulation was reinforced not only by the change in pain stimulus intensity, but also by delivery of non-painful electrical stimulation, labelled as TENS stimulation.

Questionnaire-based treatment expectation ratings were collected before conditioning and after conditioning, and after the test session, which provided an explicit measure of participant's expectations about the manipulation.

The finding that placebo and nocebo effects are influenced by recent experience provides a novel insight into a potential moderator of individual placebo effects.

Weaknesses:

There are a limited number of trials per test condition (10) which means that the trajectory of responses to the manipulation may not be explored, which would be an interesting future study.

The differences between the nocebo and control condition in pain ratings during conditioning could be explained by differing physiological effects of the different stimulus intensities, so it is difficult to make any claims about the expectation effects here. A randomisation error meant that 25 participants received an unbalanced number 448 of trials per condition (i.e., 10 x VAS 40, 14 x VAS 60, 12 x VAS 80), although the authors accounted for this during analysis so it is not of major concern.

This manuscript presents a study on expectation manipulation to induce placebo and nocebo effects in healthy participants. The study follows standard placebo experiment conventions with use of TENS stimulation as the placebo manipulation. The authors were able to achieve their aims. A key finding is that placebo and nocebo effects were predicted by recent experience, which is a novel contribution to the literature. The findings provide insights into the differences between placebo and nocebo effects and the potential moderators of these effects.

Comments on revisions:

I am satisfied with the author's revisions to the manuscript and have no further comments.

<https://doi.org/10.7554/eLife.105753.2.sa2>

Reviewer #2 (Public review):

Summary:

Kunkel et al aim to answer a fundamental question: Do placebo and nocebo effects differ in magnitude or longevity? To address this question, they used a powerful within-participants design, with a very large sample size (n=104), in which they compared placebo and nocebo effects - within the same individuals - across verbal expectations, conditioning, testing phase, and a 1-week follow-up. With elegant analyses, they establish that different mechanisms underlie the learning of placebo vs nocebo effects, with the latter being acquired faster and extinguished slower. This is an important finding for both the basic understanding of learning mechanisms in humans and for potential clinical applications to improve human health.

Strengths:

Beyond the above - the paper is well-written and very clear. It lays out nicely the need for the current investigation and what implications it holds. The design is elegant, and the analyses are rich, thoughtful, and interesting. The sample size is large which is highly appreciated, considering the longitudinal, in-lab study design. The question is super important and well-investigated, and the entire manuscript is very thoughtful with analyses closely examining the underlying mechanisms of placebo versus nocebo effects.

Comments on revisions:

The authors have addressed all of my concerns and comments - one point for them to verify is that indeed analyses that have not been preregistered will be flagged as such. The provided pre-registration link doesn't specify much about the analysis plans and specific tests used.

<https://doi.org/10.7554/eLife.105753.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

This manuscript presents a study on expectation manipulation to induce placebo and nocebo effects in healthy participants. The study follows standard placebo experiment conventions with the use of TENS stimulation as the placebo manipulation. The authors were able to achieve their aims. A key finding is that placebo and nocebo effects were predicted by recent experience, which is a novel contribution to the literature. The findings provide insights into the differences between placebo and nocebo effects and the potential moderators of these effects.

Specifically, the study aimed to:

- (1) assess the magnitude of placebo and nocebo effects immediately after induction through verbal instructions and conditioning*
- (2) examine the persistence of these effects one week later, and*
- (3) identify predictors of sustained placebo and nocebo responses over time.*

Strengths:

An innovation was to use sham TENS stimulation as the expectation manipulation. This expectation manipulation was reinforced not only by the change in pain stimulus intensity, but also by delivery of non-painful electrical stimulation, labelled as TENS stimulation.

Questionnaire-based treatment expectation ratings were collected before conditioning and after conditioning, and after the test session, which provided an explicit measure of participants' expectations about the manipulation.

The finding that placebo and nocebo effects are influenced by recent experience provides a novel insight into a potential moderator of individual placebo effects.

We thank the reviewer for their thorough evaluation of our manuscript and for highlighting the novelty and originality of our study.

Weaknesses:

There are a limited number of trials per test condition (10), which means that the trajectory of responses to the manipulation may not be adequately explored.

We appreciate the reviewer's comment regarding the number of trials in the test phase. The trial number was chosen to ensure comparability with previous studies addressing similar research questions with similar designs (e.g. Colloca et al., 2010). Our primary objective was to directly compare placebo and nocebo effects within a within-subject design and to examine their persistence one week after the first test session. While we did not specifically aim to investigate the trajectory of responses within a single testing session, we fully agree that a comprehensive analysis of the trajectories of expectation effects on pain would be a valuable extension of our work. We have now acknowledged this limitation and future direction in the revised manuscript.

The paragraph reads as follows: "It is important to note that our study was designed in alignment with previous studies addressing similar questions (e.g., Colloca et al., 2010). Our primary aim was to directly compare placebo and nocebo effects in a within-subject design and assess their persistence of these effects one week following the first test session. One limitation of our approach is the relatively short duration of each session, which may have limited our ability to examine the trajectory of responses within a single session. Future studies could address this limitation by increasing the number of trials for a more comprehensive analysis."

On day 8, one stimulus per stimulation intensity (i.e., VAS 40, 60, and 80) was applied before the start of the test session to re-familiarise participants with the thermal stimulation. There is a potential risk of revealing the manipulation to participants during the re-familiarization process, as they were not previously briefed to expect the painful stimulus intensity to vary without the application of sham TENS stimulation.

We thank the reviewer for the opportunity to clarify this point. Participants were informed at the beginning of the experiment that we would use different stimulation intensities to re-familiarize them with the stimuli before the second test session. We are therefore confident that participants perceived this step as part of a recalibration rather than associating it with the experimental manipulation. We have added this information to the revised version of the manuscript.

The paragraph now reads as follows: "On day 8, one stimulus per stimulation intensity (i.e., VAS 40, 60 and 80) was applied before the start of the test session to re-familiarise participants with the thermal stimulation. Note that participants were informed that these pre-test stimuli were part of the recalibration and refamiliarization procedure conducted prior to the second test session."

The differences between the nocebo and control conditions in pain ratings during conditioning could be explained by the differing physiological effects of the different stimulus intensities, so it is difficult to make any claims about expectation effects here.

We appreciate the reviewer's comment and agree that, despite the careful calibration of the three pain stimuli, we cannot entirely rule out the possibility that temporal dynamics during the conditioning session were influenced by differential physiological effects of the varying stimulus intensities (e.g., intensity-dependent habituation or sensitization). We have

addressed this in the revision of the manuscript, but we would like to emphasize that the stronger placebo effects during the test phase are statistically controlled for any differences in the conditioning session.

The paragraph now reads: “This asymmetry is noteworthy in and of itself because it occurred despite the equidistant stimulus calibration relative to the control condition prior to conditioning. It may be the result of different physiological effects of the stimuli over time or amplified learning in the placebo condition, consistent with its heightened biological relevance, but it could also be a stronger effect of the verbal instructions in this condition.”

A randomisation error meant that 25 participants received an unbalanced number of 448 trials per condition (i.e., 10 x VAS 40, 14 x VAS 60, 12 x VAS 80).

We agree that this is indeed unfortunate. However, we would like to point out that all analyses reported in the manuscript have been controlled for the VAS ratings in the conditioning session, i.e., potential effects of the conditioned placebo and placebo stimuli. Moreover, we have now conducted additional analyses, presented here in our response to the reviewers, to demonstrate that this imbalance did not systematically bias the results. Importantly, the key findings observed during the test phase remain robust despite this issue.

Specifically, when excluding these 25 participants from the analyses, the reported stronger placebo compared to placebo effects in the test session on day 1 remain unchanged. Likewise, the comparison of placebo and placebo effects between days 1 and 8 shows the same pattern when excluding the participants in question. The only exception is the interaction between effect (placebo vs placebo) x session (day 1 vs day 8), which changed from a borderline significant result ($p = .049$) to insignificant ($p = .24$). However, post hoc tests continued to show the same pattern as originally reported: a significant reduction in the placebo effect from day 1 to day 8 and no significant change in the placebo effect.

Reviewer #2 (Public review):

Summary:

Kunkel et al aim to answer a fundamental question: Do placebo and placebo effects differ in magnitude or longevity? To address this question, they used a powerful within-participants design, with a very large sample size ($n=104$), in which they compared placebo and placebo effects - within the same individuals - across verbal expectations, conditioning, testing phase, and a 1-week follow-up. With elegant analyses, they establish that different mechanisms underlie the learning of placebo vs placebo effects, with the latter being acquired faster and extinguished slower. This is an important finding for both the basic understanding of learning mechanisms in humans and for potential clinical applications to improve human health.

Strengths:

Beyond the above - the paper is well-written and very clear. It lays out nicely the need for the current investigation and what implications it holds. The design is elegant, and the analyses are rich, thoughtful, and interesting. The sample size is large which is highly appreciated, considering the longitudinal, in-lab study design. The question is super important and well-investigated, and the entire manuscript is very thoughtful with analyses closely examining the underlying mechanisms of placebo versus placebo effects.

We thank the reviewer for their positive evaluation of our manuscript and for acknowledging the methodological rigor and the significant implications for clinical applications and the broader research field.

Weaknesses:

There were two highly addressable weaknesses in my opinion:

(1) I could not find the preregistration - this is crucial to verify what analyses the authors have committed to prior to writing the manuscript. Please provide a link leading directly to the preregistration - searching for the specified number in the suggested website yielded no results.

We thank the reviewer for pointing this out. We included a link to the preregistration in the revised manuscript. This study was pre-registered with the German Clinical Trial Register (registration number: DRKS00029228; <https://drks.de/search/de/trial/DRKS00029228>).

(2) There is a recurring issue which is easy to address: because the Methods are located after the Results, many of the constructs used, analyses conducted, and even the main placebo and nocebo inductions are unclear, making it hard to appreciate the results in full. I recommend finding a way to detail at the beginning of the results section how placebo and nocebo effects have been induced. While my background means I am familiar with these methods, other readers will lack that knowledge. Even a short paragraph or a figure (like Figure 4) could help clarify the results substantially. For example, a significant portion of the results is devoted to the conditioning part of the experiment, while it is unknown which part was involved (e.g., were temperatures lowered/increased in all trials or only in the beginning).

We thank the reviewer for their helpful comment and agree that the Results section requires additional information that would typically be provided by the Methods section if it directly followed the Introduction. In response, we have moved the former Figure 4 from the Methods section to the beginning of the Results section as a new Figure 1, to improve clarity. Further, we have revised the Methods section to explicitly state that all trials during the conditioning phase were manipulated in the same way.

Recommendations for the Authors:

Reviewer #1 (Recommendations for the authors):

(1) Given that the authors are claiming (correctly) that there is only limited work comparing placebo/nocebo effects, there are some papers missing from their citations:

Nocebo responses are stronger than placebo responses after subliminal pain conditioning - - Jensen, K., Kirsch, I., Odmalm, S., Kaptchuk, T. J. & Ingvar, M. Classical conditioning of analgesic and hyperalgesic pain responses without conscious awareness. Proc. Natl. Acad. Sci. USA 112, 7863-7 (2015)

We thank the reviewer and have now included this relevant publication into the introduction of the revised manuscript.

Hird, E.J., Charalambous, C., El-Deredy, W. et al. Boundary effects of expectation in human pain perception. Sci Rep 9, 9443 (2019). <https://doi.org/10.1038/s41598-019-45811-x>

We thank the reviewer for suggesting this relevant publication. We have now included it into the discussion of the revised manuscript by adding the following paragraph:

“Recent work using a predictive coding framework further suggests that nocebo effects may be less susceptible to prediction error than placebo effects (Hird et al., 2019), which could

contribute to their greater persistence and strength in our study.”

(2) The trial-by-trial pain ratings could have been usefully modelled with a computational model, such as a Bayesian model (this is especially pertinent given the reference to Bayesian processing in the discussion). A multilevel model could also be used to increase the power of the analysis. This is a tentative suggestion, as I appreciate it would require a significant investment of time and work - alternatively, the authors could acknowledge it in the Discussion as a useful future avenue for investigation, if this is preferred.

We thank the reviewer for this thoughtful suggestion. While we agree that computational modelling approaches could provide valuable insights into individual learning, our study was not designed with this in mind and the relatively small number of trials per condition and the absence of trial-by-trial expectancy ratings limit the applicability of such models. We have therefore chosen not to pursue such analysis but highlight it in the discussion as a promising direction for future research.

“Notably, the most recent experience was the most predictive in all three analyses; for instance, the placebo effect on day 8 was predicted by the placebo effect on day 1, not by the initial conditioning. This finding supports the Bayesian inference framework, where recent experiences are weighted more heavily in the process of model updating because they are more likely to reflect the current state of the environment, providing the most relevant and immediate information needed to guide future actions and predictions²⁴. Interestingly, while a change in pain predicted subsequent nocebo effects, it seemed less influential than for placebo effects. This aligns with findings that longer conditioning enhanced placebo effects, while it did not affect nocebo responses¹⁰ and the conclusion that nocebo instruction may be sufficient to trigger nocebo responses. Using Bayesian modeling, future studies could identify individual differences in the development of placebo and nocebo effects by integrating prior experiences and sensory inputs, providing a probabilistic framework for understanding the underlying mechanisms.”

(3) The paper is missing any justification of sample size, i.e. power analysis - please include this.

We apologize for the missing information on our a priori power analysis. As there is a lack of prior studies investigating within-subjects comparisons of placebo and nocebo effects that could inform precise effect size estimates for our research question, we based our calculation on the ability to detect small effects. Specifically, the study was powered to detect effect sizes in the range of $d = 0.2 - 0.25$ with $\alpha = .05$ and power = .9, yielding a required sample size of $N = 83-129$. We have now added this information to the methods section of the revised manuscript.

(4) "On day 8, one stimulus per stimulation intensity (i.e., VAS 40, 60 and 80) was applied before the start of the test session to re-familiarise participants with the thermal stimulation."

What were the instructions about this? Was it before the electrode was applied? This runs the risk of unblinding participants, as they only expect to feel changes in stimulus intensity due to the TENS stimulation.

We thank the reviewer for pointing out the potential risk of unblinding participants due to the re-familiarization process prior to the second test session. We would like to clarify that we followed specific procedures to prevent participants from associating this process with the experimental manipulation. The re-familiarisation with the thermal stimuli was conducted after the electrode had been applied and re-tested to ensure that both stimulus modalities were re-introduced in a consistent and neutral context. Participants were explicitly informed

that both procedures were standard checks prior to the actual test session (“We will check both once again before we begin the actual measurement.”). For the thermal stimuli, we informed participants that they would experience three different intensities to allow the skin to acclimate (e.g., “...we will test the heat stimuli in 3 trials with different temperatures, allowing your skin to acclimate to the stimuli. ...”), without implying any connection to the experimental conditions.

Importantly, this re-familiarization procedure mirrored what participants had already experienced during the initial calibration session on day 1. We therefore assume that participants interpreted as a routine technical step rather than part of the experimental manipulation. We have now clarified this procedure in the methods section of the revised manuscript.

(5) "For a comparison of pain intensity ratings between time-points, an ANOVA with the within-subject factors Condition (placebo, nocebo, control) and Session (day 1, day 8) was carried out. For the comparison of placebo and nocebo effects between the two test days, an ANOVA with the with-subject factors Effect (placebo effect, nocebo effect) and Session (day 1, day 8) was used."

It seems that one ANOVA is looking at raw pain scores and one is looking at difference scores, but this is a bit confusing - please rephrase/clarify this, and explain why it is useful to include both.

We thank the reviewer for highlighting this point. Our primary analyses focus on placebo and nocebo effects, which we define as the difference in pain intensity ratings between the control and the placebo condition (placebo effect) and the nocebo and the control condition (nocebo effect), respectively.

To examine whether condition effects were present at each time-point, we first conducted two separate repeated measures ANOVAs - one for day 1 and one for day 8 - with the within-subject factor CONDITION (placebo, nocebo, control).

To compare the magnitude and persistence of placebo and nocebo effects over time, we then calculated the above-mentioned difference scores and submitted these to a second ANOVA with within-subject factors EFFECT (placebo vs. nocebo effect) and SESSION (day 1 vs. day 8). We have now clarified this approach on page 19 of the revised manuscript. To avoid confusion, the Condition x Session ANOVA has been removed from the manuscript.

(6) Please can the authors provide a figure illustrating trial-by-trial ratings during test trials as well as during conditioning trials?

In response to the reviewer's point, we now provide the trial-by-trial ratings of the test phases on days 1 and 8 as an additional figure in the Supplement (Figure S1) and would like to clarify that trial-by-trial pain intensity ratings of the conditioning phase are displayed in Figure 2C of the manuscript,

(7) "Separate multiple linear regression analyses were performed to examine the influence of expectations (GEE ratings) and experienced effects (VAS ratings) on subsequent placebo and nocebo effects. For day 1, the placebo effect was entered as the dependent variable and the following variables as potential predictors: (i) expected improvement with placebo before conditioning, (ii) placebo effect during conditioning and (iii) the expected improvement with placebo before the test session at day 1"

The term "placebo effect during conditioning" is a bit confusing - I believe this is just the effect of varying stimulus intensities - please could the authors be more explicit on the terminology they use to describe this? NB changes in pain rating during the conditioning

trials do not count as a placebo/nocebo effect, as most of the change in rating will reflect differences in stimulation intensity.

We agree with the reviewer that the cited paragraph refers to the actual application of lower or higher pain stimuli during the conditioning session, rather than genuinely induced placebo or nocebo effect. We thank the reviewer for this helpful observation and have revised the terminology, accordingly, now referring to these as “pain relief during conditioning” and “pain worsening during conditioning”.

(8) Supplementary materials: "The three temperature levels were perceived as significantly different (VAS ratings; placebo condition: M= 32.90, SD= 16.17; nocebo condition: M= 56.62, SD= 17.09; control condition: M= 80.84, SD= 12.18"

This suggests that the VAS rating for the control condition was higher than for the nocebo condition. Please could the authors clarify/correct this?

We thank the reviewer for spotting this error. The values for the control and the nocebo condition had accidentally been swapped. This has now been corrected in the manuscript: control condition: M= 56.62, SD= 17.09; nocebo condition: M= 80.84, SD= 12.18.

(9) "To predict placebo responses a week later (VAScontrol - VASplacebo at day 8), the same independent variables were entered as for day 1 but with the following additional variables (i) the placebo effect at day 1 and (ii) the expected improvement with placebo before the test session at day 8."

Here it would be much clearer to say 'pain ratings during test trials at day 1'.

We agree with the reviewer and have revised the manuscript as suggested.

(10) For completeness, please present the pain intensity ratings during conditioning as well as calibration/test trials in the figure.

Please see our answer to comment (6).

(11) In Figure 1a, it looks like some participants had rated the control condition as zero by day 8. If so, it's inappropriate to include these participants in the analysis if they are not responding to the stimulus. Were these the participants who were excluded due to pain insensitivity?

On day 8, the lowest pain intensity ratings observed were VAS 3 in the placebo condition and VAS 2 in the control condition, both from the same participant. All other participants reported minimum values of VAS 11 or higher (all on a scale from 0-100). Thus, no participant provided a pain rating of VAS 0, and all ratings indicated some level of pain perception in response to the stimulus. We did not define an exclusion criterion based on day 8 pain ratings in our preregistration, and we did not observe any technical issues with the stimulation procedure. To avoid post-hoc exclusions and maintain consistency with our preregistered analysis plan, we therefore decided to include all participants in the analysis.

(12) "Comparison of day 1 and day 8. A direct comparison of placebo and nocebo effects on day 1 and day 8 pain intensity ratings showed a main effect of Effect with a stronger nocebo effect ($F(1,97)= 53.93$, $131 p< .001$, $\eta^2= .36$) but no main effect of Day ($F(1,97)= 2.94$, $p= .089$, $\eta^2 = .029$). The significant Effect x Session interaction indicated that the placebo effect and the nocebo effect developed differently over time ($F(1,97)= 3.98$, $p= .049$, $\eta^2 = .039$)"

This is confusing as it talks about a main effect of "day" and then interaction with "session" - are they two different models? The authors need to clarify.

We thank the reviewer for pointing this out. In our analysis, "Session" is the correct term for the experimental factor, which has two factor levels, "day 1" and "day 8". This has now been corrected in the revised manuscript.

Reviewer #2 (Recommendations for the authors):

(1) More information on how "size of the effect" in Figures 1b and 2b was calculated is needed; this can be in the legend. If these are differences between control and each condition, then they were reversed for one condition (nocebo?), which is ok - but this should be clearly explained.

We agree with the reviewer and have now revised the figure legends to improve clarity. The legends now read:

1b: "Figure 1. Pain intensity ratings and placebo and nocebo effects during calibration and test sessions. (A) Mean pain intensity ratings in the placebo, nocebo and control condition during calibration, and during the test sessions at day 1 and day 8. (B) Placebo effect (control condition - placebo condition, i.e., positive value of difference) and nocebo effect (nocebo condition - control condition, i.e., positive value of difference) on day 1 and day 8. Error bars indicate the standard error of the mean, circles indicate mean ratings of individual participants. ***: $p < .001$, **: $p < .01$, n.s.: non-significant."

2b: "Figure 2. Mean and trial-by-trial pain intensity ratings, placebo and nocebo effects during conditioning. (A) Mean pain intensity ratings of the placebo, nocebo and control condition during conditioning. (B) Placebo effect (control condition - placebo condition, i.e., positive value of difference) and nocebo effect (nocebo condition - control condition, i.e., positive value of difference) during conditioning. (C) Trial-by-trial pain intensity ratings (with confidence intervals) during conditioning. Error bars indicate the standard error of the mean, circles indicate mean ratings of individual participants. ***: $p < .001$."

(2) In the methods, I was missing a clear understanding of how many trials there were in the conditioning phase, and then how many in the other testing phases. Also, how long did the experiment last in total?

We apologize that the exact number of trials in the testing phases was not clear in the original manuscript. We now indicate on page 18 of the revised manuscript that we used 10 trials per condition in the test sessions. We have also added information on the duration of each test day (i.e., three hours on day 1 and one hour on day 8) on page 15.

(3) In expectancy ratings, line 186 - are improvement and worsening expectations different from expected pain relief? It is implied that these are two different constructs - it would be helpful to clarify that.

We agree that this is indeed confusing and would like to clarify that both refer to the same construct. We used the Generic rating scale for previous treatment experiences, treatment expectations, and treatment effects (GEEE questionnaire, Rief et al. 2021) that discriminates between expected symptom improvement, expected symptom worsening, and expected side effects due to a treatment. We now use the terms "expected pain relief" and "expected pain worsening" throughout the whole manuscript.

(4) In the last section of the Results, somatosensory amplification comes out of nowhere - and could be better introduced (see point 2 above).

We agree with the reviewer that introducing the concept of somatosensory amplification and its potential link to placebo/nocebo effects only in the Methods is unhelpful, given that this section appears at the end of the manuscript. We therefore now introduce the relevant publication (Doering et al., 2015) before reporting our findings on this concept.

(5) In line 169, if the authors want to specify what portion of the variance was explained by expectancy, they could conduct a hierarchical regression, where they first look at R² without the expectancy entered, and only then enter it to obtain the R² change.

We fully agree that hierarchical regression can be a useful approach for isolating the contribution of variables. However, in our case, expectancy was assessed at different time points (e.g., before conditioning and before the test session on day 1), and there was no principled rationale for determining the order in which these different expectancy-related variables should be entered into a hierarchical model.

That said, in response to the reviewer's suggestion, we have now conducted hierarchical regression analyses in which all expectancy-related variables were entered together as a single block (see below). These analyses largely confirmed the findings reported so far and are provided here in the response to the reviewers below. Given the exploratory nature of this grouping and the lack of an a priori hierarchy, we feel that the standard multiple regression models remain the most appropriate for addressing our research question because it allows us to evaluate the total contribution of expectancy-related predictors while also examining the individual contribution of each variable within the block. We would therefore prefer to retain these as the primary analyses in the manuscript.

Results of the hierarchical regression analyses:

Day 1 - Placebo response: In step 1, we entered the difference in pain intensity ratings between the control and the placebo condition during conditioning as a predictor. In step 2, we added the two variables reflecting expectations (i.e., expected improvement with placebo (i) before conditioning and (ii) before the test session on day 1). This allowed us to assess whether expectation-related variables explained additional variance beyond the effect of conditioning.

The overall regression model at step 1 was significant, $F(1, 102) = 13.42$, $p < .001$, explaining 11.6% of the variance in the dependent variable ($R^2 = .116$). Adding the expectancy-related predictors in step 2 did not lead to a significant increase in explained variance, $\Delta R^2 = .007$, $F(2, 100) = 0.384$, $p = .682$. Thus, the conditioning response significantly predicted placebo-related pain reduction on day 1, but additional information on expectations did not account for further variance.

Day 1 - Nocebo response: The equivalent analysis was run for the nocebo response on day 1. In step 1, the pain intensity difference between the nocebo and the control condition was entered as a predictor before adding the two expectancy ratings (i.e., expected worsening with nocebo (i) before conditioning and (ii) before the test session on day 1).

In step 1, the regression model was not statistically significant, $F(1, 102) = 2.63$, $p = .108$, and explained only 2.5% of the variance in nocebo response ($R^2 = .025$). Adding the expectation-related predictors in Step 2 slightly increased the explained variance by $\Delta R^2 = .027$, but this change was also non-significant, $F(2, 100) = 1.41$, $p = .250$. The overall variance explained by the full model remained low ($R^2 = .052$). These results suggest that neither conditioning nor expectation-related variables reliably predicted nocebo-related pain increases on day 1.

Day 8 - Placebo response: For the prediction of the placebo effect on day 8, the following variables reflecting perceived effects were entered as predictors in step 1: the difference in pain intensity ratings between the control and the placebo condition (i) during conditioning and (ii) on day 1. In step 2, the variables reflecting expectations were added: the expected improvement with placebo (i) before conditioning, (ii) before the test session on day 1 and (iii) before the test session on day 8.

In step 1, the model was statistically significant, $F(3, 95) = 14.86$, $p < .001$, explaining 23.8% of the variance in the placebo response ($R^2 = .238$, Adjusted $R^2 = .222$). In step 2, the addition of the expectation-related predictors resulted in a non-significant improvement in model fit, $\Delta R^2 = .051$, $F(3, 92) = 2.21$, $p = .092$. The overall variance explained by the full model increased modestly to 29.0%.

Day 8 - Nocebo response: For the equivalent analyses of nocebo responses on day 8, the following variables were included in step 1: the difference in pain intensity ratings between the nocebo and the control condition (i) during conditioning and (ii) on day 1. In step 2, we entered the variables reflecting nocebo expectations including expected worsening with nocebo (i) before conditioning, (ii) before the test session on day 1 and (iii) before the test session on day 8. In step 1, the model significantly predicted the day 8 nocebo response, $F(3, 95) = 6.04$, $p = .003$, accounting for 11.3% of the variance ($R^2 = .113$, Adjusted $R^2 = .094$). However, the addition of expectation-related predictors in Step 2 resulted in only a negligible and non-significant improvement, $\Delta R^2 = .006$, $F(3, 92) = 0.215$, $p = .886$. The full model explained just 11.9% of the variance ($R^2 = .119$).

Typos:

(6) Abstract - 104 heathy xxx (word missing).

(7) Line 61 - reduce or decrease - I think you meant increase.

Thank you, we have now corrected both sentences.

References

Colloca L, Petrovic P, Wager TD, Ingvar M, Benedetti F. How the number of learning trials affects placebo and nocebo responses. *Pain*. 2010

Doering BK, Nestoriuc Y, Barsky AJ, Glaesmer H, Brähler E, Rief W. Is somatosensory amplification a risk factor for an increased report of side effects? Reference data from the German general population. *J Psychosom Res*. 2015

<https://doi.org/10.7554/eLife.105753.2.sa0>